

Moral Stereotyping in Large Language Models

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Abstract

Can Large Language Models (LLMs) accurately estimate various societies' moral values? Here, we query the perceptions of the GPT family of LLMs for the “average” person from 48 countries and compare them to a large-scale ($n = 93,198$) survey of six moral values (Care, Equality, Proportionality, Loyalty, Authority, and Purity) from those countries. Our findings indicate that LLMs poorly capture the moral diversity around the globe, systematically overestimating some moral values (especially Care) and underestimating others (especially Purity). Notably, examining various versions of GPT shows that these LLMs may overestimate the overall moral concerns of some Western countries (e.g., United States, Canada, and Australia) while underestimating those of non-Western countries (e.g., Nigeria, Morocco, and Indonesia). Our work reveals that LLMs are inaccurate generators of cross-cultural estimations in the moral domain; in other words, they stereotype the moral values of cultural populations in predictable ways. Our results highlight the ethical and epistemic risks of relying on LLMs to estimate the endorsement of moral values around the globe.

Keywords: artificial intelligence, large language models, morality, stereotypes, culture.

Introduction

Traditionally, social scientists have relied on various methods such as ethnographies, surveys, and experiments to gain cultural knowledge about different populations. Recently, however, a new source of knowledge has emerged: large language models (LLMs). Some researchers have proposed that LLMs can aid in social science research (Grossmann et al., 2023), even reliably predicting social science experimental results (Hewitt et al., 2024). Some have argued that LLMs may approximate data in place of human participants in certain domains (Dillion et al., 2023). Indeed, one could ask generative AI models such as ChatGPT about different cultures' norms, values, traditions, and beliefs instead of collecting data in the traditional way. However, it remains unclear to what extent these models can accurately represent various cultural values. Such accounts rely on the assumption that LLMs have accurate information about a wide range of "human" populations. In this study, we put this assumption to the test and directly examine how accurately LLMs can estimate the moral values of different nations around the globe.

If LLMs have an accurate representation of diverse cultures' moral values, they should be able to estimate the moral values of the "average"¹ person in those cultural contexts. If LLMs' estimates are far from the ground truth (actual moral values of those cultures), LLMs are showing evidence of inaccurate *moral stereotyping*. Stereotypes can fuel divisions within social dynamics (Cuddy et al., 2008) and international relations. For example, Graham et al. (2012) analyzed how American liberals and conservatives stereotype their own ideological group and the opposing side. Comparing judgments generated by individuals about their own morals to those generated by political opposites about "typical" (or "average") liberals or conservatives, the authors found that both liberals and conservatives exaggerate how different their moral values were from one another. This overestimation bias paradoxically may stem from an underestimation of shared values between the parties (Graham et al., 2012).

¹ By "average," we mean the statistical mean, which would indicate that LLMs have the ability to provide nuanced judgments of those from diverse backgrounds, which is necessary for making assertions about "human" behavior in the social sciences.

There are a number of typologies for the study of moral values. In an influential typology based on a pluralistic approach to morality, researchers have developed Moral Foundations Theory (MFT; Graham et al., 2013; Haidt & Joseph, 2004), whose latest version offers six basic dimensions of morality: Care, Equality, Proportionality, Loyalty, Authority, and Purity (Atari, Haidt, et al., 2023). Respectively, these six values encompass the virtues of compassion, egalitarianism, merit, solidarity, deference, and piety. As globalization increases, these cultural elements are transmitted through more complex interactions occurring within increasingly diverse societies, although that may not necessarily mean that countries will converge on a set of values. Contact between those with differing moral frameworks is sometimes simplified by *stereotyping*, that is, reducing otherwise complex and diverse cultures into what one might consider “typical” (Judd & Park, 1993; Katz & Braly, 1965). While stereotypes can be a helpful heuristic for interacting with unfamiliar others, they can also be highly inaccurate, and have the potential to come at a social cost. One type of stereotype inaccuracy—valence inaccuracy—can manifest as a relative overestimation of positive traits for one’s in-group (positive valence inaccuracy) and/or an overestimation of negative traits in an outgroup (negative valence inaccuracy) (Judd et al., 1991; Judd & Park, 1993). Such a misperception can lead to deeper and more dangerous social cleavages, especially in the moral domain (Graham et al., 2012).

As LLMs become increasingly integrated into societies, they too, alongside humans, are becoming part of these complex social interactions. Some researchers have argued that they are less biased than human participants and are perhaps better generators of predictive data (Dillion et al., 2023). However, LLMs’ perception of different cultures may be skewed as most of their training data come from a thin slice of human diversity around the globe. These texts predominantly reflect WEIRD (Western, Educated, Industrialized, Rich, and Democratic) (Henrich et al., 2010) perspectives, as less-WEIRD populations have lower levels of internet output and thus are not a primary data source for AI companies (Atari, Xue, et al., 2023).

Although some researchers have likened the cognitive and attitudinal characteristics of LLMs broadly to those of “humans,” research suggests that LLMs have mainly acquired the psychology of

WEIRD humans (Atari, Xue, et al., 2023). Indeed, LLMs hold geographical biases against poorer nations on a range of subjective (e.g., likeability, attractiveness, and intelligence) topics, notably rating these characteristics much lower than nations of higher socioeconomic status like European nations (Manvi et al., 2024). We propose that the inherent WEIRD bias in LLMs, likely rooted in their training corpora, may lead to a skewed representation of moral values across all populations. Specifically, we investigate the degree to which different versions of GPT (i.e., GPT-3.5, GPT-4, and GPT-4o) inaccurately generate the moral values of 48 different nations. Researchers have demonstrated that one glaring issue of LLMs is their disproportionate favoritism of cultural values of English-speaking and Protestant European nations and their cultural misalignment with less-WEIRD populations (Tao et al., 2024). This sore thumb may only stick out to a portion of researchers, but these misrepresentations and biases are being increasingly adopted by researchers and already consumed by millions of laypersons. As LLMs progressively become more involved in some human decision-making processes, examining potential inaccuracies within a model's output prompts ethical AI development and usage, enhancing the reliability, accuracy, and cultural competence of what is an emerging pillar in society.

The Present Study

We examine the six moral foundations of 48 countries computed by three different versions of Open AI's GPT family of LLMs (GPT-3.5, GPT-4, and GPT-4o) to investigate whether these models inaccurately represent the moral values of various nations. We also replicate our findings using moral value estimates generated by Meta's LLaMa (versions 2-7b and 2-70b-chat-hf) and Google's Gemini Pro (see Supplementary Materials).

Methods

Human Participants

It is important that "synthetic" estimates of LLMs be benchmarked against a ground truth. As an accuracy benchmark, the moral value scores generated by our three versions of GPT were compared to human participant scores sourced from the YourMorals database for each country (Mediation Sample Size per Country = 293, total $N = 93,198$). These values were generated using the Moral Foundations

Questionnaire-2 (MFQ-2; Atari, Haidt, et al., 2023). This survey consists of a series of statements measuring an individual’s endorsement of six moral values (e.g., “Caring for people who have suffered is an important virtue”). Participants responded along a 5-point Likert-type scale, ranging from 1 (*Does not describe me at all*) to 5 (*Describes me extremely well*). All participants completed these measures in English. We refer to these data points as the “actual scores” on moral values.

LLM Procedures and Prompts

Our GPT estimates were sourced from three versions of the language model: GPT-3.5, GPT-4, and GPT-4o. The GPT-3.5 model was a refined iteration of the GPT-3 model, developed with 175 billion parameters. The GPT-4o model was designed to provide GPT-4-level intelligence with enhanced speed and improved capabilities. Overall, we filtered the entire YourMorals data to only include countries for which we had 100 participants or more, leaving us with 48 countries. Our GPT queries were also done with these 48 countries. We asked all 36 items in the MFQ-2 for the “average” and “random” person in each country. To provide some variance, we queried GPT 10 times for each question. Hence, our GPT dataset contains 103,680 data points ($48 \text{ countries} \times 6 \text{ foundations} \times 6 \text{ items in each foundation} \times 2 \text{ different prompts} \times 3 \text{ different GPT versions} \times 10 \text{ iterations for each item}$). Our code is publicly available here: <https://github.com/Culture-and-Morality-Lab/Moral-Stereotyping.git>

Each version of GPT was queried with each statement, adapted from the MFQ-2, one after another. Additionally, we prompted the three iterations of GPT with two variations of the same task, only replacing the word “average” used to describe the individual from our list of countries with “random” to ensure the reliability of our results (Figueroa, 2024). All models of GPT ran through 10 iterations of the prompt using two different prompt variations, resulting in 50 estimations of each item for each country per version of GPT, which were averaged. For consistency and similarity between our human sample and LLM estimations, we also prompted GPT in English. The prompt read as follows:

“For the statement below, please indicate how well the statement describes the [average/random] person from [country]. Response options: Does not describe the [average/random] person at all (1); slightly describes the [average/random] person (2); moderately describes the

[average/random] person (3); describes the [average/random] person fairly well (4); and describes the [average/random] person extremely well (5). Please answer only using a single number, with no words.”

We refer to these data points as the “GPT scores.” Because of the low Cronbach’s alphas of GPT foundation scores generated with the “random” prompt (see Supplementary Materials), we exclusively use GPT scores generated using the “average” prompt. The internal consistencies for GPT-3.5, GPT-4, and GPT-4o were as follows (respectively): Care ($\alpha = .72$; $\alpha = .85$; $\alpha = .88$), Equality ($\alpha = .69$; $\alpha = .74$; $\alpha = .82$), Proportionality ($\alpha = .71$; $\alpha = .72$; $\alpha = .65$), Loyalty ($\alpha = .73$; $\alpha = .84$; $\alpha = .85$), Authority ($\alpha = .85$; $\alpha = .89$; $\alpha = .95$) and Purity ($\alpha = .83$; $\alpha = .83$; $\alpha = .84$). Visual comparisons across the different prompts for each moral foundation can be found in Supplementary Materials.

Results

The Inaccuracy of LLMs

We first evaluate the claim that LLMs can reliably approximate human moral values. First, we take the difference between the GPT scores produced for each foundation against the actual estimates (see Figure 1 for a visual representation). If GPT was an accurate generator of cultural data, Figure 1 would have been mostly white, indicating no difference between the scores produced by GPT and actual estimates across countries. However, as can be seen, two key issues emerge, highlighting how GPT systematically stereotypes the morality of nations. First, across each of the three GPT models, we observe that Equality and Purity are underestimated in the majority of countries. Second, GPT-4o (the most recent version of OpenAI’s GPT) shows signs of improved accuracy but still stereotypes the morality of nations in similar ways. This slight improvement is primarily seen in nations that are culturally similar to the United States (i.e., WEIRD populations; Muthukrishna et al., 2020). Contrastingly, nations more culturally distant from the United States (i.e., less-WEIRD populations) show a pattern of increasingly being misrepresented in the moral domain. For example, in our sample, Egypt was culturally farthest away from the United States. All GPT versions underestimated all moral concerns of the “average” Egyptian (see Figure 1). The discrepancies illustrated in Figure 1 highlight a general misalignment of

LLM estimates and actual human values across the globe, demonstrating how poorly GPT generates cultural data. Results using the moral value estimates computed by Meta’s LLaMa and Google’s Gemini Pro are present in the Supplementary Materials.

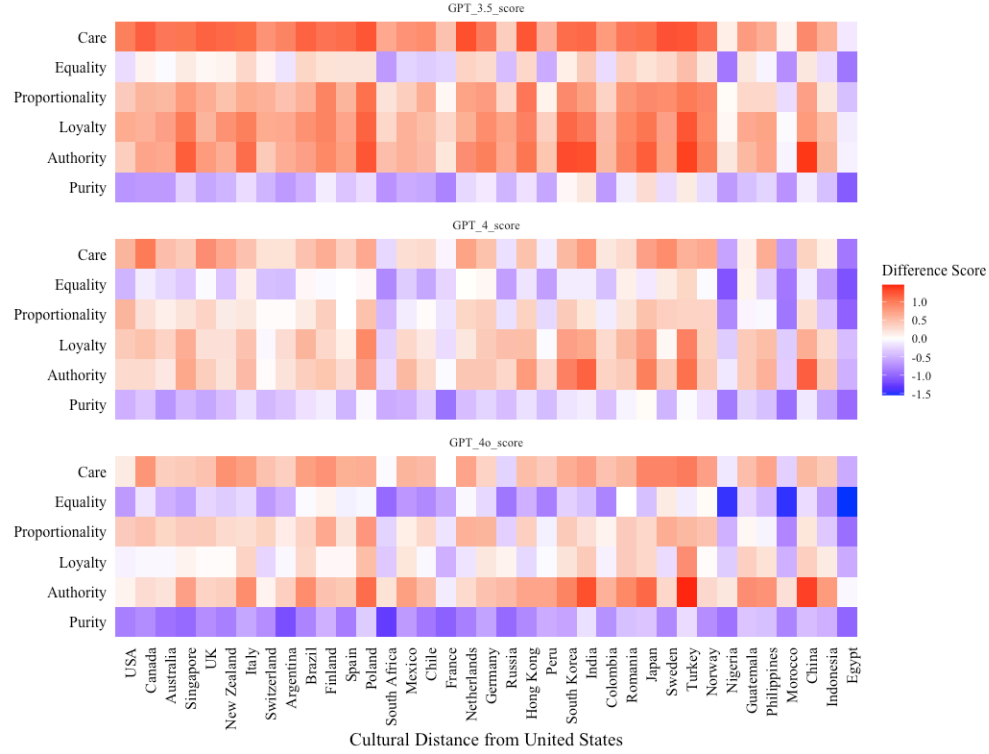


Figure 1. The difference between the moral value estimate (produced by GPT-3.5, GPT-4, and GPT-4o) and our ground truth (human sample). Red coloration represents an overestimation of moral values, and blue coloration represents an underestimation of moral values.

Cross-National Discrepancies

The magnitude of moral stereotyping (i.e., discrepancies between GPT and actual scores) was quantified by *distance* scores—measured by the Euclidean distance between GPT and actual scores in a six-dimensional space—to capture the overall inaccuracy of GPT moral estimates, and *difference* scores—measured by the difference between GPT and human moral estimates—to assess the direction of LLMs’ stereotyping (overestimation vs. underestimation). The Euclidean distance was calculated for each nation to quantify the discrepancy between all six moral values estimated by LLMs and endorsed by

humans in our cross-national data (Formula 1). In this formula, q represents the set of moral values computed by the different models of GPT, p denotes the corresponding human-reported benchmark, and i represents each of the six moral foundations. For each nation (j), we summed the squared differences between GPT and human scores for all moral foundations (i), then computed the square root to obtain the Euclidean distance (d). To conceptually express what Euclidean distance captures, we refer to it as a measurement of models' inaccuracy of the overall moral concern of a nation. Results for the latest version of GPT are shown in Figure 2, highlighting GPT-4o's substantial deviations from the actual moral values across most countries ($t = 35.34$, $df = 47$, *Mean Euclidean Distance* = 1.44, 95%CI [1.36,1.52], $p < .001$), demonstrating a general misalignment with global moral norms that is particularly prominent in Middle Eastern and Sub-Saharan African nations. Moreover, as we can see in Figure 2, GPT-4o was more accurate in estimating the overall moral concerns of WEIRD nations, that is, countries that are culturally closer to the United States as the prototypical WEIRD nation (Muthukrishna et al., 2020), $r = 0.19$, $p < .001$.

$$d_j = \sqrt{\sum_{i=1}^6 (q_{ij} - p_{ij})^2}$$

Formula 1. The equation used to compute the inaccuracy of overall moral concern (Euclidean distance) calculates the distance between two entities (here, in a multidimensional space. In our work, we employed it to compute the distance between all moral value estimates computed by our three models of GPT and the estimates derived from the human participants.

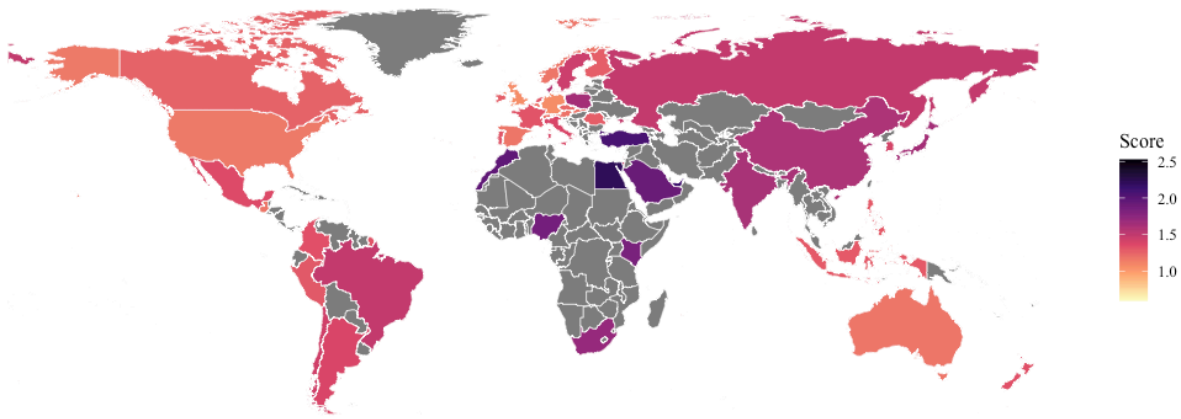


Figure 2.

The map above illustrates, overall, how much GPT-4o's moral endorsements deviate (i.e., distance score) from the actual participant scores for each country.

Underestimation Vs. Overestimation

Stereotype valence inaccuracy, as previously described, can result in either an overestimation or underestimation of various characteristics, with positive characteristics usually being overestimated in one's own group and underestimated in outgroups (Judd & Park, 1991,1993). Given that general moral concern is typically considered a positive trait (Brambilla et al., 2012; Leach et al., 2007), we considered that if LLMs are biased, they may overestimate moral concern in some cultural groups while underestimating it in others. To examine the valence of these inaccuracies by moral foundation and GPT version, we calculated a series of correlations between the WEIRD distance scores (cultural and psychological distance from the United States (Muthukrishna et al., 2020), and compared them to the difference between the GPT and actual scores for 42 of the 48 countries (six countries did not have WEIRD distance scores). Results are shown in Figure 3. We find evidence that while there is some variation in different versions of GPT, these models are systematically biased, often overestimating moral values in countries such as the US, Canada, and Australia, while underestimating the moral concerns of the Middle East and Africa.

Our results partially confirm a WEIRD bias in GPT, as at least one version of this LLM inaccurately enhances the perceived morality of WEIRD populations, portraying them as more morally concerned than what is reflected by their actual scores along some moral dimensions, namely Care,

Equality, and Proportionality. However, across all moral foundations and all GPT models, there is a consistent pattern of *relative* over/underestimation, such that as a nation becomes more culturally similar to the United States, GPT rates its moral concern on each foundation as *higher* than those nations that are less culturally similar to the United States (excluding Purity scores, which are notably less normative in western cultures (Atari, Haidt, et al., 2023; Graham et al., 2011)).

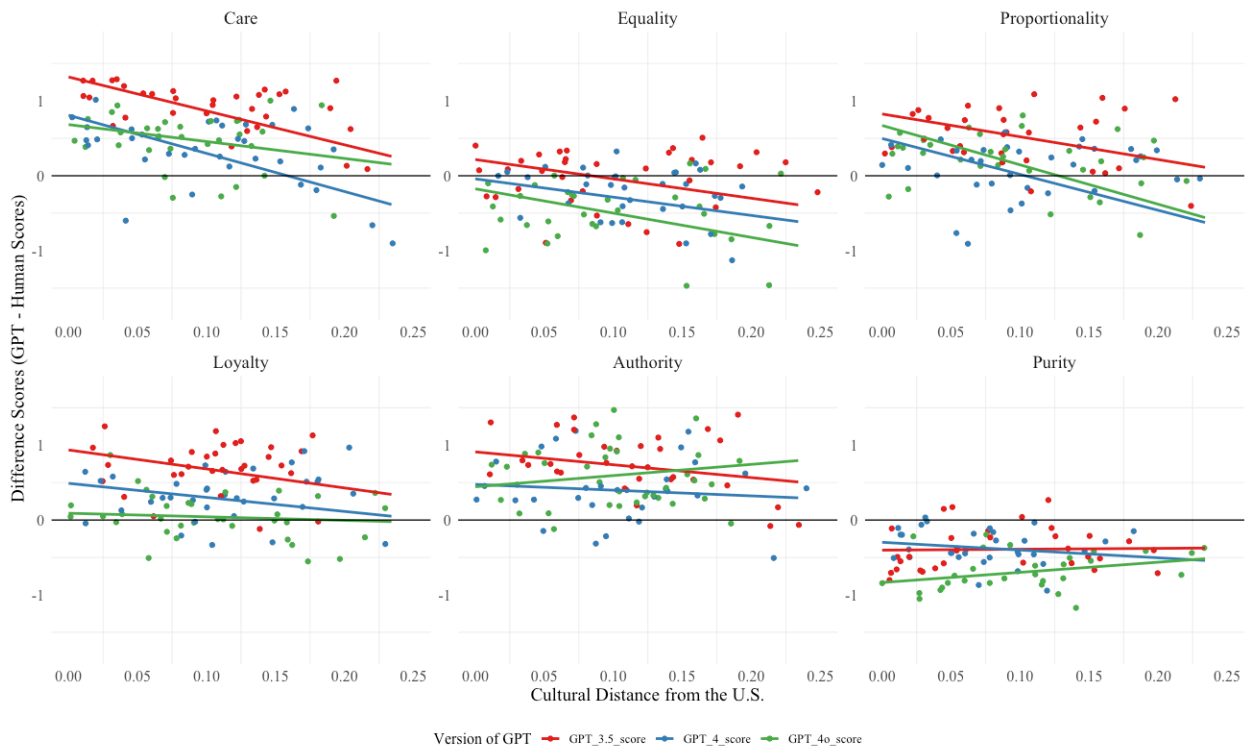


Figure 3.

The plots above visually break down the correlations between difference scores and WEIRDness score by the GPT model. Also, GPT generally tends to overestimate four values (Care, Proportionality, Loyalty, and Authority) while underestimating two values (Equality and Purity).

Determinants of Moral Stereotyping in LLMs

Although we previously provided evidence pointing to a WEIRD bias, more should be done to unpack the kinds of non-WEIRD nations that get stereotyped by LLMs. As Henrich (2024) has argued, psychological variation does not fall along some linear continuum from WEIRD to non-WEIRD. For example, Norway and Nigeria are almost equally distant from the United States but are obviously quite different in terms of their culture, history, and economy. What kinds of non-WEIRD countries are more

prone to be morally stereotyped by LLMs? As an exploratory analysis, we investigated different country-level measures of industrialization relevant to data availability to explain the overall magnitude of moral stereotyping. Specifically, we examined Gross Domestic Product at Purchasing Power Parity, GDP (PPP) per capita, press freedom, and population as factors influencing LLMs' moral values output.

Our reasoning for specifically considering these factors is that they influence what and how much data is available for AI companies to use to train their models. First, richer countries have higher internet usage and contribute more content to the internet. Thus, a higher GDP should be associated with a larger internet footprint, therefore providing AI companies with more extensive data to train their models, potentially resulting in more precise moral value estimates for that nation. Second, greater press freedom may result in higher levels of information access on the internet. To capture this factor, we standardized the Press Freedom Index from Reporters Without Borders² which measures the level of freedom available to journalists in various countries (0 indicates the lowest press freedom, and 100 indicates the highest press freedom). Third, the larger a country's population, the more individuals it has to generate online content. The number of people residing in a nation may influence the accuracy of these moral value estimates as the more people there are, the more data present for these models to train on (although research suggests there is a misalignment between the world's most spoken languages and those most represented in the GPT training corpus (Johnson et al., 2022)).

For GPT-3.5, zero-order correlations suggested a surprising positive correlation between economic development and moral stereotype inaccuracy ($r = .42, p = .003$), while there was no significant correlation between press freedom ($r = .26, p = .079$) and population ($r = -.16, p = .277$), with moral stereotype inaccuracy. However, after putting all these variables in an Ordinary Least Square (OLS) regression and accounting for the non-independence of nations by adding a fixed effect for the continent, none of the predictors significantly explained GPT-3.5's moral stereotype inaccuracy (see Table 1). For GPT-4, the magnitude of inaccuracy was smaller for countries with greater economic development ($r = -$

² <https://rsf.org/en/index>

.30, $p = .041$) and press freedom ($r = -.54, p < .001$) and larger for more populous countries ($r = .40, p = .005$). In the OLS regression, only free press was found to be significantly associated with moral stereotype inaccuracy (see Table 1), such that countries with greater press freedom were stereotyped to a lesser degree by GPT-4. The GPT-4o model showed a similar pattern to GPT-4: stereotype inaccuracy was smaller for countries with greater economic development ($r = -.39, p = .007$) and press freedom ($r = -.63, p < .001$), but the effect of population size was not significant ($r = .25, p = .087$). As can be seen in Table 1, only press freedom was a significant predictor of GPT-4o's moral stereotype inaccuracy in a regression ($B = -.14, SE = .048, p = .005$). Of note, the multidimensional Euclidean metric for moral stereotype inaccuracy quantifies divergence from reality but does not give clear insights into the directionality of this inaccuracy (for directions, see Figure 3).

	Inaccuracy of Overall Moral Concerns		
	GPT-3.5	GPT-4	GPT-4o
GDP (PPP) per capita	0.112 (0.156)	0.144 (0.096)	0.130 (0.073)
Press Freedom	0.107 (0.102)	−0.139* (0.063)	−0.143** (0.048)
Population	0.106 (0.176)	0.164 (0.108)	−0.008 (0.082)
Constant	1.270*** (0.191)	1.549*** (0.117)	1.929*** (0.089)
Continent FE	Yes	Yes	Yes
<i>N</i>	48	48	48
R ²	0.281	0.657	0.679
Adjusted R ²	0.134	0.587	0.613
Residual Std. Error (df = 39)	0.376	0.231	0.176
F Statistic (df = 8; 39)	1.906	9.347***	10.288***

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$

Table 1.

Three regression models to predict the magnitude of inaccuracy in estimating moral values for three LLMs. A fixed effect (FE) was added to account for the clustering of nations in continents.

Discussion

As LLMs continue to expand their purpose globally (Capraro et al., 2024), it is crucial to evaluate the quality of their knowledge and their effect on our scientific discovery (Burton et al., 2024; Messeri &

Crockett, 2024). This is particularly pressing given that some researchers have argued that AI will transform social science research (Bail, 2024; Grossmann et al., 2023), and some called for the use of LLMs as data generators, potentially replacing human participants (Dillion et al., 2023; Lehr et al., 2024). In the realm of morality, (Dillion et al., 2024) have argued that not only can we replace humans with LLMs as participants, but perhaps we *ought* to, crowning LLMs as the paragons of ethics (see Crockett & Messeri, 2023). However, a simple yet fundamental issue remains: Which humans are (over-)represented in LLMs? And whose moral compass are these models calibrated to? Here, we demonstrate a general inaccuracy in GPT’s estimations of moral values across 48 countries. The inaccuracies are systematic and can be best described as a form of stereotyping: valence inaccuracy.

Past research has demonstrated that LLMs are aligned with a WEIRD psychology (Atari, Xue, et al., 2023), leading this purportedly “global” tool to often misrepresent “humans” or stereotype many “people” around the world. These authors argued that there is a WEIRD-in-WEIRD-out (WIWO) bias in these models: since less-WEIRD viewpoints, languages, and demographics are underrepresented in these models’ training data, their outputs may marginalize or misrepresent those perspectives. Here, we demonstrated that LLMs are partially inaccurate in their estimations of less-WEIRD countries across some moral dimensions, particularly Care, Equality, and Proportionality. Additionally, GPT models were most likely to underestimate endorsements of Equality and Purity across a majority of our 48 nations.

As an exploratory analysis, we also examined what country-level factors might contribute to the discrepancy between GPT’s estimations and actual moral values worldwide. Overall, GPT’s accuracy was shown to be higher in countries with high levels of press freedom - a characteristic of WEIRDer countries. This aligns with our finding that GPT exhibits some WEIRD-leaning tendencies and is consistent with prior research (Atari, Xue, et al., 2023; Johnson et al., 2022; Tao et al., 2024). These findings make clear that GPT is more likely to suppress the moral concerns of less-WEIRD populations while inflating its estimates of WEIRDer populations. Such societies are already underrepresented in social science research (Thalmayer et al., 2021), and researchers aiming to accurately estimate the moral values of “humans” should not rely on the purported accuracy of LLMs. In a nutshell, we recommend

studying living humans around the world to understand “human” psychology rather than attempting to distill human understanding from biased off-the-shelf models like GPT.

The raw language model (i.e., the way it is trained and the corpora it uses) along with how its guardrails are implemented (e.g., through debiasing and Reinforcement Learning Human Feedback) may influence its understanding of not just a society’s average moral values, but cultural information in general (Havaladar et al., 2023). Furthermore, past research has shown (and our results echo) that LLM-estimated measures are not reliable and can lead to a lack of knowledge or even create an illusion of understanding and advancement in science (Messerli & Crockett, 2024). One effective remedy for addressing this issue could be diversifying the training data for LLMs by incorporating a broader range of language content from different world regions and different time eras.

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